Technical Document #46: Cybersecurity - Comprehensive Overview

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Encryption converts plaintext into ciphertext to protect data confidentiality. AES and RSA are widely used encryption algorithms.

Zero trust architecture assumes no implicit trust and verifies every request. It minimizes the attack surface and limits lateral movement.

Penetration testing simulates cyber attacks to identify vulnerabilities. It helps organizations assess their security posture before real attacks occur.

Social engineering manipulates people into revealing confidential information. Phishing is the most common social engineering attack.

The rapid advancement of technology has transformed how organizations approach problem-solving and innovation. Companies must adapt to stay competitive in an ever-evolving landscape.

Sustainability and environmental responsibility are becoming key considerations in technology design. Green computing aims to reduce the environmental impact of technology.

Augmented reality overlays digital information on the physical world. It has applications in training, maintenance, and entertainment.

Artificial intelligence continues to reshape industries from healthcare to finance. Machine learning models can now perform tasks that were once thought to require human intelligence.

Cloud computing has democratized access to powerful computing resources. Startups can now access the same infrastructure as large enterprises.

Computer vision applications are becoming ubiquitous, from facial recognition to autonomous driving. Deep learning has enabled breakthroughs in visual understanding.

Autonomous vehicles use a combination of sensors, AI, and mapping to navigate without human intervention. Self-driving technology is rapidly advancing.

Key Takeaways:

- Security information and event management (SIEM) systems collect and analyze security logs.
- Intrusion detection systems (IDS) monitor network traffic for suspicious activity.
- Penetration testing simulates cyber attacks to identify vulnerabilities.